

Pythagorean Triplet

Spencer Veatch

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Problem 0.1 – Special Pythagorean Triplet A Pythagorean Triplet is a set of three natural numbers $a < b < c$, for which:

$$a^2 + b^2 = c^2$$

For example, $3^2 + 4^2 = 9 + 16 = 25 = 5^2$. There exists exactly one Pythagorean Triplet for which $a + b + c = 1000$. Find the product abc .

0.1 Reducing The Search Space

Since this problem gives both a sum constraint and a Pythagorean constraint, we can reduce the search space algebraically. Formally, the problem asks us to find the Pythagorean Triplet (a, b, c) satisfying the system:

$$a^2 + b^2 = c^2$$

$$a + b + c = 1000$$

Given the ordering constraint $a < b < c$. Using the sum constraint, we can solve for c :

$$c = 1000 - a - b$$

which can be substituted into the Pythagorean Equation as

$$a^2 + b^2 = (1000 - a - b)^2$$

$$a^2 + b^2 = (1000000 + a^2 + b^2 - 2000a - 2000b + 2ab)$$

$$0 = 1000000 - 2000a - 2000b + 2ab$$

$$0 = 500000 - 1000a - 1000b + ab$$

This representation can be rearranged and factored with respect to b as:

$$ab - 1000a - 1000b = -500000$$

$$b(a - 1000) = 1000a - 500000$$

$$b = \frac{1000a - 500000}{(a - 1000)}$$

Thus, for each admissible value of a , the corresponding value of b can be uniquely determined.

0.2 Bounding the Domain

Given the inequality

$$a < b < c$$

and the knowledge that $a + b + c = 1000$, it follows that:

$$\begin{aligned} 3a &< 1000 \\ a &< \frac{1000}{3} \end{aligned}$$

Ergo, the search is reduced to the finite interval:

$$[1 \leq a \leq 333].$$

The algorithmic solution must simply iterate over all possible values of a , computing the corresponding values of b and c , then verifying whether $a^2 + b^2 = c^2$ holds.